4.6 Find and fix

In this lesson

You will learn:

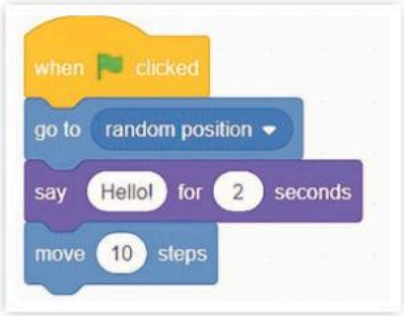
- how to make changes to a program
- how to find and fix errors.

Commands to match actions

A student wanted to make a program. They decided the sprite would:

- jump to a random place
- say "Hello!"
- move around the screen forever.

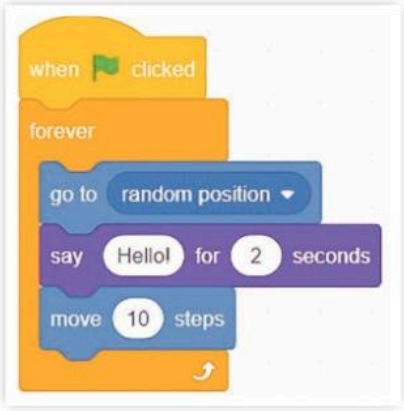
The student found blocks to match the actions. Here is the program made of those blocks.



The student ran the program. The program did not work the way they wanted. The sprite moved just once, then stopped.

Using a loop

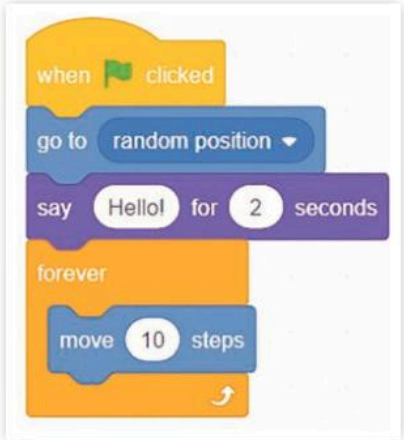
The student decided to use a 'forever loop'. They put all the command blocks inside the loop. Here is the program they made.



The student ran the program. The program did not work the way they wanted. Too many blocks are inside the loop. The sprite keeps jumping to a new place. The sprite keeps saying "Hello!"

Inside and outside the loop

Some commands should go before the loop. Those are the commands that must happen only once.



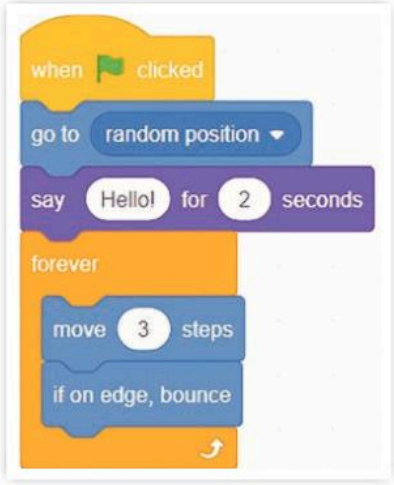
Some commands should be put inside the loop. Those are the commands that will be repeated.

Look at the final picture on the previous page. You will see the program with all the commands in the right place.

Add some bounce

The sprite moves until it is almost off the screen and then stops. You must add a block that tells the sprite to 'bounce' if it hits the side.

The finished program is shown here.





Activity

Using the ideas on this page make a program that makes a butterfly move about the screen.



Extra challenge

Look in the program you made to find the block that says 'move 10 steps'.

Change the number. Run the program and see what is different.

Explore what happens when you use different numbers.



Explore more

Make a new program. Use a sprite that looks like a ball. Make it bounce around the screen.

Add different types of ball. Set them to different speeds.



Check what you know

You have learned

- how a program is made of commands
- how to pick commands to make the program you want
- how to make a program that works
- how to make changes and corrections to a program.

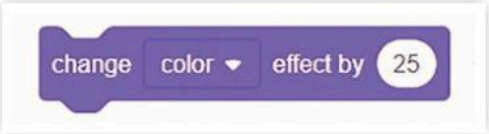
Test



Here is a program made using Scratch.
Pick one answer for each question.

1. What does this program do?
   - **a** Make the sprite move about.
   - **b** Click the green flag.
   - **c** Play a game.
2. What happens at the start of the program?
   - **a** The sprite moves forward 10 steps.
   - **b** The sprite says "Hello!"
   - **c** The sprite goes to a random position.
3. What commands will be repeated?
   - **a** The sprite asks your name and says "Hello!" to you.
   - **b** The sprite points towards the mouse and moves forward.
   - **c** The sprite goes to a random place and waits 1 second.

4 If you add this block inside the forever loop what will happen?



a The sprite changes colour once.

b The sprite changes colour lots of times.

c The sprite changes colour to red.



Activities

1 Make a program that does this:

- The sprite walks forward 10 steps.
- The sprite says "Goodbye!"
- The sprite walks another 10 steps.

2 Plan and make a program that has a forever loop in it. You can decide what commands to put in your program.

Self-evaluation

- I answered test question 1 and started activity 1.
- I answered test questions 1–3. I completed activity 1.
- I answered all the test questions. I completed both activities.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?